Two-Dimensional Array Questions

1. Take a matrix of R x C order and find sum of boundary elements
2. Take a matrix of M x M size, find sum of both the diagonals and check if they are equal
3. Take a matrix of R x C order and print its transpose
4. Take a matrix or order M x M and check if it's symmetric or not. A matrix is symmetric if $A=A^T$ (A transpose)
5. Take a matrix of order M x M and check if it's skew-symmetric matrix or not. A matrix is skew symmetric if $A^T=-A$
6. Take a matrix of order R x C and print sum of each row and each column.
7. Take a matrix of order R x C and find largest and smallest element of the array
8. Take a matrix of size R x C and check if sum of all upper primary diagonal elements are equal to the sum of all lower primary diagonal elements or not
9. Take a matrix of size R x C and check if sum of all upper secondary diagonal elements are equal to the sum of all lower secondary diagonal elements or not
10. Take a matrix of order R x C and find sum of all corner elements.
11. Take a matrix of order R x C. sort its elements in ascending order
12. Take two matrices of order R x C and find their addition and subtraction
13. Initialize the following matrix and find its mirror matrix

5	10	15
20	25	30
35	40	45

14. Take a square matrix of order M x M and check if it is an **identity matrix** or not.
    (An identity matrix has 1s on the diagonal and 0s elsewhere)
15. Take a matrix of order R x C and check if it's a **sparse matrix**.
    (A matrix is sparse if the number of 0s is more than the number of non-zero elements)
16. Take a matrix of order R x C and count **how many prime numbers** are present in the matrix. Also print the prime numbers
17. Take a matrix of order R x C and display the **row number with the maximum sum**.
18. Take a matrix of order R x C and check if the matrix is a **magic square**.
    (A magic square has the same sum for all rows, columns, and both diagonals)
19. Take a matrix of order R x C and **swap the first and last rows**.
20. Take a square matrix of order M x M and **check if it's a diagonal matrix**.
    (All non-diagonal elements should be 0)